

ICEIS 2026 - Paper #112

Paper Title: *Harder, Better, Faster, Stronger: Benchmarking Search Algorithm Efficiency in Complex Mazes*

Reviewer #1

General Assessment (Please assign scores using the following criteria (1=weakest; 6=strongest))

Relevance (Paper fits one or more of the topic areas?): 3

Originality (Newness of the ideas expressed): 1

Technical Quality (Theoretical soundness/methodology): 2

Significance (Is the problem worth the given attention?): 2

Presentation (Structure/Length/English): 2

Overall Rating (Weighted value of above items): 2

Improvement Suggestions (for authors to consider in the camera-ready version. Additional detail in "Observations")

Abstract and Introduction are adequate? Yes

Needs more experimental results? Yes

Needs comparative evaluation? Yes

Improve critical discussion ? (validation): Yes

Figures are adequate ? (in number and quality): No

Conclusions/Future Work are convincing? No

References are up-to-date and appropriate? No

Paper formatting needs adjustment? Yes

Improve English? Yes

Detailed comments to authors, including aspects that must be improved in the camera-ready version of the paper:

The paper presents an empirical evaluation of the BFS and DFS algorithms for autonomous robot navigation in finding a target object within a maze. No proper justification for the empirical analysis is given, especially considering that the properties of such classical algorithms are already known. The experimental setup and the investigation can be regarded as a didactic exercise, yielding no meaningful results from a scientific standpoint. The adopted execution plan is somewhat questionable, as in each scenario (as per fig. 1), only one single instance of the maze is used; instead, multiple topologies should be considered to get to more general results.

The descriptive text is approximate, lacking clarity. The listed equations make no sense without a formal definition of the terms. No abstract view of the experiment, apart from a mere list of actions taken, is given.

The quality of figures is low. The description of fig 3 is not coherent with what is shown.

Regarding the list of references, it could be improved by inserting more recent (and grounded) works, carried out also in 2025 – it's just a matter of quickly searching on Internet.

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Paper Title: *Harder, Better, Faster, Stronger: Benchmarking Search Algorithm Efficiency in Complex Mazes*

Reviewer #2

General Assessment (Please assign scores using the following criteria (1=weakest; 6=strongest))

Relevance (Paper fits one or more of the topic areas?): 4
Originality (Newness of the ideas expressed): 2
Technical Quality (Theoretical soundness/methodology): 3
Significance (Is the problem worth the given attention?): 3
Presentation (Structure/Length/English): 4
Overall Rating (Weighted value of above items): 3

Improvement Suggestions (for authors to consider in the camera-ready version. Additional detail in "Observations")

Abstract and Introduction are adequate? Yes
Needs more experimental results? Yes
Needs comparative evaluation? Yes
Improve critical discussion ? (validation): Yes
Figures are adequate ? (in number and quality): Yes
Conclusions/Future Work are convincing? No
References are up-to-date and appropriate? Yes
Paper formatting needs adjustment? No
Improve English? Yes

Detailed comments to authors, including aspects that must be improved in the camera-ready version of the paper:

The paper addresses a relevant and well-established problem in mobile robotics, namely the comparative performance of BFS and DFS in maze-like environments during exploration phases. The topic is appropriate for the scope of the conference and the experimental setup is generally sound from an implementation perspective. However, the scientific contribution remains limited, as the work does not introduce any novel algorithmic ideas, theoretical insights, or methodological advances beyond what is already well documented in the literature. The comparison between BFS and DFS, even in the specific context of exploration with unknown target positions, is largely incremental and confirms known trade-offs between memory usage, traversal strategy, and computational cost.

The experimental evaluation, while clearly described, is restricted to a simulated environment with a limited set of maze configurations and performance metrics. The analysis would benefit from a stronger methodological framing, including statistical validation of results, discussion of variance and robustness across multiple random seeds, and a more explicit justification of why the chosen experimental setting provides new insights compared to existing benchmark studies. Moreover, the discussion of results tends to be mostly descriptive, and the interpretation of why DFS outperforms BFS in the reported scenarios remains at an intuitive level, without deeper theoretical or empirical grounding.

To strengthen the contribution, the authors should consider extending the experimental campaign to include additional baseline algorithms (e.g., heuristic search methods), a broader set of environments, and a clearer articulation of the practical implications of the findings for real robotic systems beyond simulation. The conclusions and future work section would also benefit from a more critical reflection on the limitations of the study and a clearer positioning of the results within the broader research landscape. Overall, while the paper is technically correct and clearly written, its contribution is currently too limited and incremental to justify acceptance in its present form.